

1) $P_0(-3, 0, 7)$ 'den geçen $\vec{n} = 5\hat{i} + 2\hat{j} - \hat{k}$ 'ya dik düzlem?

$$(x+3)\hat{i} + (y-0)\hat{j} + (z-7)\hat{k} \perp 5\hat{i} + 2\hat{j} - \hat{k}$$

$$5(x+3) + 2y + (z-7) = 0$$

$$5x + 2y - z + 22 = 0$$

2) $P_0(2, 1, 4)$ noktasından geçen $\ell: \begin{cases} x=2+t \\ y=1+2t \\ z=3 \end{cases}$ doğrusuna dik düzlem?

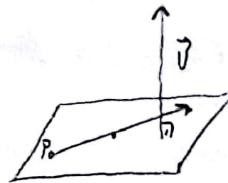
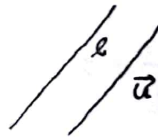
$$\frac{x-2}{1} = \frac{y-1}{2} = \frac{z-3}{0} = t$$

$$\vec{u} = \hat{i} + 2\hat{j}$$

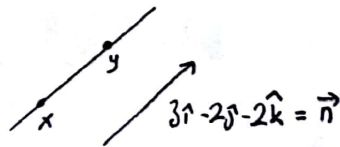
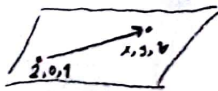
$$(x-2)\hat{i} + (y-1)\hat{j} + (z-4)\hat{k} \perp \hat{i} + 2\hat{j}$$

$$(x-2) + 2(y-1) = 0$$

$$x + 2y - 4 = 0$$



3) $(2, 0, 1)$ 'den geçen ve $x(1, 1, 0)$, $y(4, -1, -2)$ noktalarından geçen doğruya dik düzlem?



$$3(x-2) + (-2y) - 2(z-1) = 0$$

$$3x - 2y - 2z - 4 = 0$$

4) $2x+3y+3z=6$ düzleminin normalini, düzlem üzerinde bir noktası ve eksenlerin kesim noktası?

$$\vec{n} = 2\hat{i} + 3\hat{j} + 3\hat{k}$$

$$x=y=1, z=1/3$$

$$x=0 \dots$$

5) $x-2y+5z-1=0$ düzlemi ile $\vec{r}(t) = \langle 2-t, 1+2t, t-1 \rangle$ doğrusu arasındaki ilişki?

$$\frac{x-2}{-1} = \frac{y-1}{2} = \frac{z+1}{1} = t \rightarrow \text{doğrunun kanonik denklemi}$$

$$-\hat{i} + 2\hat{j} + \hat{k} = \vec{v} \rightarrow \text{doğruya paralel vektör.}$$

$$x-2y+5z-1=0 \text{ düzleminin normali } \vec{n} = \hat{i} - 2\hat{j} + 5\hat{k}$$

$$\vec{n} \cdot \vec{v} = 0, \text{ paraleller.}$$

6) $3x-2y+z=2$ ve $x-y+3z=8$ düzlemlerinin arakesit doğrusunun parametrik denklemi?

$$\vec{n}_1 = 3\hat{i} - 2\hat{j} + \hat{k}$$

$$\vec{n}_2 = \hat{i} - \hat{j} + 3\hat{k}$$

$$\begin{aligned} \vec{n}_3 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -2 & 1 \\ 1 & -1 & 3 \end{vmatrix} = \hat{i} \begin{vmatrix} -2 & 1 \\ -1 & 3 \end{vmatrix} - \hat{j} \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix} + \hat{k} \begin{vmatrix} 3 & -2 \\ 1 & -1 \end{vmatrix} \\ &= -5\hat{i} - 8\hat{j} - \hat{k} \end{aligned}$$

arakesit üzerinde nokta $\rightarrow x=1, y=2, z=3$

$$\frac{x-1}{-5} = \frac{y-2}{-8} = \frac{z-3}{-1} = t$$

$$L: \begin{cases} x=1-5t \\ y=2-8t \\ z=3-t \end{cases}$$

$x = \frac{8}{3} + 2t$, $y = -2t$, $z = 1+t$ doğrusunun $3x + 2y - 6z - 6 = 0$ düzlemini ile kesiştiği nokta?

$$3\left(\frac{8}{3} + 2t\right) + 2(-2t) + (1+t) \cdot -6 - 6 = 0$$

$$t = -1$$

$$x = \frac{2}{3}, \quad y = 2, \quad z = 0$$

8) $\vec{r}(t) = 3 \cdot \cos t \hat{i} + 3 \sin t \hat{j} + t^2 \hat{k} \rightarrow$ birim teğet vektörü?

$$\vec{t} = \frac{\frac{d\vec{r}}{dt}}{\left|\frac{d\vec{r}}{dt}\right|} = \frac{-3 \sin t \hat{i} + 3 \cos t \hat{j} + 2t \hat{k}}{\sqrt{9 + 4t^2}}$$

9) $x + \lambda y + 2z = 3$ ve $\lambda x + y - 2z = 1$ düzlemlerinin paralel olması için $\lambda = ?$

$$\vec{n}_1 = \hat{i} + \lambda \hat{j} + 2\hat{k}$$

$$\vec{n}_2 = \lambda \hat{i} + \hat{j} - 2\hat{k}$$

$$\vec{n}_1 \parallel \vec{n}_2$$

$$\hat{i} + \lambda \hat{j} + 2\hat{k} \parallel (\lambda \hat{i} + \hat{j} - 2\hat{k}) \cdot t$$

$$\frac{1}{\lambda} = \frac{\lambda}{1} = -1 = t \quad \text{ise } \lambda = -1$$

10) $\lambda x - y + 3z = 1$ ve $2x + 3y - \lambda z = 4$ düzlemlerinin dik olması için $\lambda = ?$

$$\vec{n}_1 = \lambda \hat{i} - \hat{j} + 3\hat{k}$$

$$\vec{n}_2 = 2\hat{i} + 3\hat{j} - \lambda \hat{k}$$

$$\vec{n}_1 \perp \vec{n}_2 \Rightarrow \vec{n}_1 \cdot \vec{n}_2 = 0$$

$$2\lambda - 3 - 3\lambda = 0$$

$$-\lambda = -3$$

11) $x-y+2z=3$ ve $x-y+3z=8$ ile verilen düzlemlerin paralel olmadığını göster ve arakesit doğrusunu parametrize et.

$$\vec{n}_1 \parallel \vec{n}_2 ?$$

$$\vec{n}_1 = \hat{i} - \hat{j} + 2\hat{k}$$

$$\vec{n}_2 = \hat{i} - \hat{j} + 3\hat{k}$$

$$\vec{n}_1 \parallel \vec{n}_2 \text{ için } \rightarrow (\hat{i} - \hat{j} + 2\hat{k}) \parallel t \cdot (\hat{i} - \hat{j} + 3\hat{k})$$

$$1 = 1 = \frac{2}{3} = t \rightarrow \text{yanlış olduğundan paralel değildir}$$

$$\vec{v} \perp (\vec{n}_1 \times \vec{n}_2)$$

$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 2 \\ 1 & -1 & 3 \end{vmatrix} = \hat{i} \begin{vmatrix} -1 & 2 \\ -1 & 3 \end{vmatrix} - \hat{j} \begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} + \hat{k} \begin{vmatrix} 1 & -1 \\ 1 & -1 \end{vmatrix} = -\hat{i} - \hat{j}$$

arakesit doğrusundan bir noktas

$$x=0$$

$$-y+2z=3$$

$$-y+3z=8$$

$$z=5$$

$$y=7$$

$$x=0$$

$$\frac{x-0}{-1} = \frac{y-7}{-1} = \frac{z-5}{0} = t \rightarrow \text{Kanonik denklem}$$

$$l_8 \begin{cases} x = -t \\ y = 7-t \\ z = 5 \end{cases}$$

$$L_1: r_1(t) = \langle 6, 1+t, 0 \rangle$$

$$L_2: r_2(s) = \langle 1, s, 0 \rangle$$

doğruların kesiştiği nokta ve aralardaki açı?

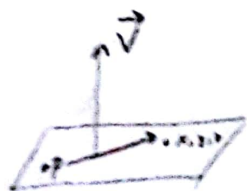
$$\begin{aligned} t=1 \\ s=2 \end{aligned} \rightarrow (1, 2, 0)$$

$$L_1: \begin{cases} x=6 \\ y=1+t \\ z=0 \end{cases} \rightarrow \frac{x-6}{1} = \frac{y-1}{1} = \frac{z-0}{0} = t \rightarrow \hat{i} + \hat{j}$$

$$L_2: \begin{cases} x=1 \\ y=s \\ z=0 \end{cases} \rightarrow \frac{x-1}{0} = \frac{y-0}{1} = \frac{z-0}{0} = s \rightarrow \hat{j}$$

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| \cdot |\vec{b}|} = \frac{1}{\sqrt{2}}, \quad \theta = \frac{\pi}{4}$$

5) $P(1,2,3)$ noktasından geçen ve $L: x=1-t, y=1+3t, z=2t$ doğrusuna dik olan düzlemin denklemi



$$L' \text{nin} \text{ kanonik} \text{ denklemi} \rightarrow \frac{x-1}{-1} = \frac{y-1}{3} = \frac{z-0}{2} = t$$

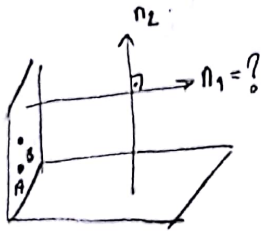
$$\text{paralel vektör} \rightarrow \vec{v} = -\hat{i} + 3\hat{j} + 2\hat{k}$$

$$(x-1)\hat{i} + (y-2)\hat{j} + (z-3)\hat{k} \perp \vec{v}$$

$$1-x+3y-6+2z-6=0$$

$$-x+3y+2z-11=0$$

14) A(1,1,1) ve B(2,0,3) noktalarından geçen ve $x+2y-3z=0$ düzlemine dik olan düzlemin denklemi = ?



$$\vec{n}_2 = \hat{i} + 2\hat{j} - 3\hat{k}$$

$$(2-1)\hat{i} + (0-1)\hat{j} + (3-1)\hat{k} \parallel (x-1)\hat{i} + (y-1)\hat{j} + (z-1)\hat{k}$$

$$\frac{x-1}{1} = \frac{y-1}{-1} = \frac{z-1}{2} = t$$

$$\vec{V} = \hat{i} - \hat{j} + 2\hat{k}$$

$$(\vec{n}_2 \times \vec{V}) = \vec{n}_1$$

$$\vec{n}_2 \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 2 \\ 1 & 2 & -3 \end{vmatrix} = -\hat{i} + 5\hat{j} + 3\hat{k} = \vec{n}_1$$

$$(x-1)\hat{i} + (y-1)\hat{j} + (z-1)\hat{k} \perp \vec{n}_1$$

$$1-x+5y-5+3z-1=0$$

15) $x-z=1$, $y+2z=3$ düzlemlerinin ortaklık doğrusunu içeren ve $x+y-2z=1$ düzlemine dik olan bir üçüncü düzlemin denklemi?

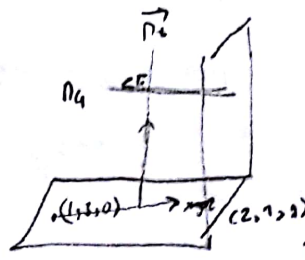
$$\vec{n}_1 = \hat{i} - \hat{k} \quad \vec{n}_2 = \hat{j} + 2\hat{k} \quad \rightarrow \quad \vec{n}_3 = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -1 \\ 0 & 1 & 2 \end{vmatrix} = \hat{i} \begin{vmatrix} 0 & -1 \\ 1 & 2 \end{vmatrix} - \hat{j} \begin{vmatrix} 1 & -1 \\ 0 & 2 \end{vmatrix} + \hat{k} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = \hat{i} - 2\hat{j} + \hat{k}$$

$$\begin{matrix} x=1, x=2 \\ y=3, y=1 \\ z=0, z=1 \end{matrix} \quad \frac{x-1}{1} = \frac{y-3}{-2} = \frac{z-0}{1} = t$$

$$\begin{cases} x=t+1 \\ y=3-2t \\ z=t \end{cases} \text{ ortaklık doğrusu}$$

$$(1, 3, 0)$$

$$x+y-2z \text{ dik}$$



$$\vec{n}_4 = \hat{i} + \hat{j} - 2\hat{k}$$

$$\hat{i} - 2\hat{j} + \hat{k} = \vec{n}_5$$

$$\vec{n}_6 = \vec{n}_4 \times \vec{n}_5 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -2 \\ 1 & -2 & 1 \end{vmatrix} = \hat{i} \begin{vmatrix} 1 & -2 \\ -2 & 1 \end{vmatrix} - \hat{j} \begin{vmatrix} 1 & -2 \\ 1 & 1 \end{vmatrix} + \hat{k} \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = -3\hat{i} - 3\hat{j} - 3\hat{k}$$

$$x+y+z-4=0$$

$$-3(x-1) - 3(y-3) - 3(z-0) = 0 \Rightarrow -3x - 3y - 3z + 12 = 0 \Rightarrow x+y+z-4=0$$